COMPUTER VISION - MIDTERM EXAM

Variant 1

Time: 120 minutes | Total: 100 points

Student name: _____ Group: _____

Student ID: _____ Date: _____

Instructions: This is an individual, handwritten, closed-book exam. Show calculations where required. Answers without explanation may receive partial credit. Phones, laptops, smart watches, notes, and AI tools are not permitted.

Question 1 - Image Processing & OpenCV [20 points]

A student is building a system to classify healthy and diseased plant leaves. The collected images have different resolutions, lighting conditions, and some contain image noise.

a) [6 pts] Propose three preprocessing operations that could be useful. For each operation, explain why you would use it.

b) [4 pts] The student converts every image to grayscale. Give one advantage and one disadvantage of doing this for plant-disease classification.

c) [5 pts] Consider the following code. Identify two problems and provide the corrected code.

```
img = cv2.imread("leaf.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)
blur = cv2.GaussianBlur(gray, (4, 4), 0)
```

d) [5 pts] A student says: "Applying more preprocessing always improves classification accuracy." Do you agree? Explain using one example.

Question 2 - Image Classification [20 points]

A classifier produces the following scores:

Image	Cat	Dog	Bird	True class
A	2.4	0.7	-0.2	Cat
B	0.3	1.2	1.8	Dog
C	-0.5	0.4	2.1	Bird
D	1.4	1.6	0.2	Dog

a) [4 pts] Write the predicted class for A, B, C, and D.

b) [4 pts] Calculate classification accuracy. Show your calculation.

c) [4 pts] Which image was incorrectly classified?

d) [4 pts] Explain the difference between a class score and a class probability.

e) [4 pts] The model achieves 95% training accuracy but 62% test accuracy. What is the likely problem? Give one possible solution.

Question 3 - Linear Classifier [25 points]

Consider a three-class linear image classifier: $s = Wx + b$

$$x = [2, -1]^T$$
$$W = [[1, 2], [-1, 1], [2, -1]]$$
$$b = [0, 1, -1]^T$$

Classes: 0 = Cat, 1 = Dog, 2 = Bird

a) [12 pts] Calculate $s = Wx + b$. Show every step.

b) [3 pts] What class does the classifier predict?

Question 3 - Linear Classifier (continued)

c) [5 pts] Explain in your own words what W and b represent.

d) [5 pts] Why can't a single linear classifier correctly separate every possible dataset? You may include a small drawing.

Question 4 - Understand the CV Pipeline [20 points]

A student wrote the following image-processing pipeline:

```
img = cv2.imread("cat.jpg")
img = cv2.resize(img, (224, 224))
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
blur = cv2.GaussianBlur(gray, (5, 5), 0)
edges = cv2.Canny(blur, 100, 200)
```

a) [5 pts] Write the dimensions of img after resizing.

b) [5 pts] What happens to the number of channels after BGR2GRAY?

c) [5 pts] Explain why Gaussian blur might be applied before Canny edge detection.

d) [5 pts] Suppose the classification task depends strongly on object color. Would you feed edges alone to the classifier? Explain your decision.

Question 5 - Design Your Own CV System [15 points]

You need to develop a system that classifies waste into Plastic | Paper | Glass | Other. You have collected 2,000 images.

a) [3 pts] How would you organize the dataset?

b) [3 pts] Propose a train/test split and calculate the number of images in each part.

Question 5 - Design Your Own CV System (continued)

c) [3 pts] Give two preprocessing operations you would consider.

d) [3 pts] What metric would you use to evaluate your classifier? Explain why.

e) [3 pts] Your model performs very well on your dataset but poorly when tested using photos taken by another student's phone. Give one possible reason.

END OF EXAM
Please check that you have answered all five questions.